SANIYA THAKUR

AI/ML Engineer | MCA Graduate

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PROFESSIONAL SUMMARY

Passionate MCA student with hands-on experience building full-stack and AI-powered applications using Python, React, and FastAPI. Skilled in developing scalable software solutions, REST APIs, and modern web applications, with familiarity in LLMs and machine learning concepts. Strong problem-solving ability with a keen interest in software engineering, full-stack development, and backend systems.

TECHNICAL EXPERTISE

- **AI & Machine Learning:** LangGraph, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Large Language Models (LLMs), Prompt Engineering, OpenCV, Hugging Face, OpenAI/Gemini APIs, Cloudflare Workers AI.
- **Languages & Core Coding:** Python, JavaScript, TypeScript, SQL, HTML, CSS.
- **Frameworks & Web UI:** React.js, FastAPI, Flask, Django, Node.js, Streamlit.
- **Databases & DevOps:** ChromaDB, MySQL, Git/GitHub, Vercel, Render, Docker, CI/CD pipelines.

PROJECTS

Multi-Modal AI Studio (Python, LangGraph, LangChain, RAG, ChromaDB, Cloudflare AI API, Streamlit)

- Architected a stateful multi-agent orchestrator using LangGraph to coordinate specialized agents (Planner, Searcher, Retriever, Writer, Critic) to execute complex research tasks with autonomous self-correction loops.
- Engineered an advanced RAG pipeline using ChromaDB and local Hugging Face embeddings to execute semantic document chunking and knowledge retrieval with zero external API overhead.
- Implemented a multi-modal media pipeline integrating Cloudflare Workers AI (FLUX.1) for ultra-fast, rate-limit-resistant image generation and Edge TTS for asynchronous audio synthes.

Live: https://multi-modal-ai-studio-ygzdttxkduvvhthpd34jbx.streamlit.app/

Multi-Modal Medical AI Agent (Python, GenAI, PaddleOCR, OpenCV, Streamlit)

- Developed a hybrid medical report analyzer combining computer vision (PaddleOCR) and generative AI (OpenRouter LLM) for intelligent text extraction and report interpretation.
- Integrated secure environment management using .env configurations to safeguard API pipelines during automated generation of structured medical summaries.
- Built a responsive web interface featuring smooth drag-and-drop document upload functionalities alongside real-time medical data extraction and abnormal findings detection.

Live: https://multi-modal-ai-agent-tqsusyxkx8so7cfgvzj9jq.streamlit.app/

Tiny Transformer Language Model (Python, PyTorch, Transformer, Streamlit)

- Built and trained a localized Transformer-based language model entirely from scratch, implementing custom tokenization, positional encoding, and multi-head self-attention.
- Optimized next-token prediction loops and deployed the model architecture behind an interactive Streamlit UI utilizing temperature-based generation sampling.

INTERNSHIPS & VIRTUAL SIMULATIONS

- Web & Application Development Intern | Binary Solutions (4 Months) – Programmed mobile UI components using Android Studio and Flutter.
- Technology Job Simulations: Data Analytics (Deloitte) | Cybersecurity Analyst (Tata) | Front-End Engineering (Skyscanner).

EDUCATION & CREDENTIALS

Master of Computer Applications (MCA) | Shivaji University, Kolhapur (Completed 2024 – 2026)
Bachelor of Computer Applications (BCA) | Punyashlok Ahilyadevi Holkar University, Solapur (Completed 2021 – 2024)